

Received 18 January 2025, accepted 1 February 2025, date of publication 7 February 2025, date of current version 12 February 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3539709

APPLIED RESEARCH

Research and Application of Digital Twin and Simulation Teaching Methodology in DVOR

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This work was supported by the Fundamental Research Funds for the Central Universities, China, under Grant J2023-014.

ABSTRACT Doppler VHF Omnidirectional Radio Range (DVOR) plays a critical role as a key component of Ground Equipment for Aircraft Navigation (GEAN). So far, little research has addressed the question of how to use Digital Twin (DT) for GEAN. Given that there is a notable paucity of empirical research specifically focusing on the DVOR data acquisition method and Virtual Reality (VR) teaching technology. The specific objective of this study is to explore the application of DT and VR teaching in DVOR. Additionally, this research specifically focuses on the VRB-52D type of the DVOR system. In this study, a novel real-time data acquisition algorithm is proposed to implement The Real-Time Data-Driven Equipment Health Monitoring Subsystem (RTDEHMS). Furthermore, fault logic rules for DVOR are proposed for fault simulation to implement The Virtual Simulation Teaching Subsystem (VSTS). This study not only addresses how to remotely monitor the operational status of DVOR but also resolves the limited opportunities for Communication, Navigation, and Surveillance Equipment Maintenance Personnel (CNS Personnel) to engage in the practical operation of DVOR during their training process. With this system, CNS Personnel can efficiently manage DVOR, and it can also help students learn about DVOR quickly. Finally, an offline survey provided quantitative data from 40 participants. The results of the survey show that almost all participants are satisfied with the system. This research provides a timely and necessary study of the DT application on the DVOR. This study not only confirms the possibility of DT system for GEAN, but also the proposed fault logic construction method is applicable to other GEAN. This study also provides practical experience in modeling, data collection, and interface standards for GEAN, facilitating the standardization and collaboration of GEAN. It provides essential technical support for modern aviation ground support.

INDEX TERMS DVOR, digital twin, virtual reality, failure simulation, aircraft navigation.

I. INTRODUCTION

Ground Equipment for Aircraft Navigation (GEAN) is a critical component of aviation systems responsible for providing navigational services to aircraft. This equipment includes Doppler VHF Omni-Directional Range (DVOR), Distance Measuring Equipment (DME), Instrument Landing System (ILS) and Non-Directional Beacon (NDB), among others. However, with the growing demand for air transport and the aging of those equipment, GEAN face several challenges, such as frequent equipment failures, high maintenance

costs, and inefficiencies in system coordination. These challenges highlight the need for innovative technological solutions to improve the reliability and efficiency of system operations. In particular, issues related to operations and maintenance management, such as equipment aging and the complexity of maintenance, pose significant obstacles. Therefore, it is crucial to research advanced GEAN operation and maintenance techniques.

In the day-to-day management of GEAN, the DVOR is a key component [1], typically operated and maintained by Communication, Navigation, and Surveillance Equipment Maintenance Personnel (CNS Personnel). The primary function of this equipment is to provide accurate navigational

The associate editor coordinating the review of this manuscript and approving it for publication was Zhenbao Liu^{ID}.

signals for aircraft positioning and guidance. While the maintenance of the equipment itself is relatively straightforward, the wide geographical distribution of this equipment presents significant challenges for unmanned management and remote monitoring. This becomes particularly critical when DVOR systems are deployed in remote areas where efficient, real-time remote management and monitoring has become a pressing issue.

Furthermore, due to the high cost of DVOR, many institutions or universities that train CNS Personnel face significant challenges in providing sufficient equipment for hands-on training. This limitation severely restricts the development of CNS Personnel skills and knowledge transfer. Therefore, exploring new technological approaches to improve the training efficiency of DVOR operators and maintenance personnel has become another key issue in current research.

This study will focus on addressing two critical issues that require urgent attention:

- (a) whether DT technology can be used for real-time monitoring and health management of equipment.
- (b) how to provide a more efficient and immersive training environment for CNS trainees.

With the rapid development of information technology, DT technology has emerged as an innovative tool that enables real-time integration of physical equipments with their digital models. This integration facilitates real-time monitoring, condition assessment and predictive maintenance of equipment. DT technology has been widely applied in industries such as manufacturing, smart cities and traffic management, and its potential for use in GEAN is becoming increasingly apparent. Through DT models, CNS Personnel can access the real-time health status of equipment, optimise maintenance strategies and significantly improve equipment availability while reducing unplanned downtime.

Meanwhile, the rapid development of VR technology has opened up new possibilities for training and operation of GEAN. VR technology allows operators to simulate equipment and conduct troubleshooting training in a virtual environment. This not only enhances the immersive training experience, but also significantly improves the training results and the mastery of operational skills, thus enabling the high-quality development of CNS Personnel.

Despite the huge potential for the application of DT and VR technologies in GEAN, several key challenges remain in both research and practice. First, ensuring efficient real-time data synchronisation and validating data integrity between the physical GEAN and its DT model remains a pressing issue. Second, enhancing interactivity, fault simulation, troubleshooting capabilities, immersion and overall training effectiveness through VR technology poses significant technical challenges. Furthermore, due to the inherent complexity and diversity of GEAN, the DT and VR systems must achieve higher levels of accuracy and flexibility in modelling and simulation to meet the requirements of different types of equipment.

The aim of this study is to explore the application of DT and VR technologies in GEAN, using the DVOR (specifically the VRB-52D model) equipment as a case study [2]. The VRB-52D model of the DVOR control panel is complex, and its model construction and equipment interaction simulation are more challenging compared to other GEAN. This research proposes an integrated DT and VR framework to improve maintenance efficiency, optimise equipment condition monitoring, improve training effectiveness, and reduce operation and maintenance costs. Specifically, the study focuses on how real-time data acquisition from DT models can be combined with immersive VR training to form a comprehensive health management and operational training system.

The contributions of this research are twofold: on the one hand, it presents an integrated solution based on DT and VR, providing innovative approaches to the maintenance and operation of GEAN, particularly in DVOR systems; on the other hand, it explores the technical pathways and implementation strategies for combining DT and VR technologies, providing new insights and directions for further research in related fields.

The structure of this paper is organized as follows: Section I introduces the importance of GEAN, presents the main research questions, objectives, and discusses the potential impact of the study. Section II reviews the literature on GEAN, discussing the application of DT technology in aviation, analyzing the role of VR in virtual teaching, and examining the combined use of DT and VR in the aviation industry. Section III presents the overall system framework design and methodology of this study. Section IV explores The Real-Time Data-Driven Equipment Health Monitoring Subsystem (RTDEHMS) through a case study of the DVOR. Section V discusses the safety issues related to GEAN. Section VI provides a detailed description of the The Virtual Simulation Teaching Subsystem (VSTS) construction for fault simulation and troubleshooting. Section VII presents the results of the questionnaire survey and provides a discussion. It explores the applicable scenarios of the system architecture and the minimum viable data required, and demonstrates the implementation of ILS's RTDEHMS using this approach, validating the system's transferability. Finally, Section VIII summarizes the key findings of this study, addressing the research questions and achieving the research objectives. It highlights the contributions and significance of the work. The section also acknowledges the limitations of the study and outlines directions for future research.

II. RELATED WORK

Given the lack of direct research on the application of DT and VR to GEAN, this section will review existing aviation studies related to GEAN, as well as research related to the application of DT and VR. Firstly, the current state of research on GEAN is discussed. This is followed by a review of existing studies on DT technology, focusing on its implementation and challenges in the aviation sector. Subsequently, the current state of VR research in educational

tasks will be investigated, along with an analysis of the potential applications of VR technology in the operation, training and simulation of GEAN, and its integration potential with DT technology. Finally, the progress of current research on the integration of DT and VR will be explored, highlighting the limitations of existing methods and identifying the innovations proposed by this study.

A. CURRENT STATUS OF GEAN RESEARCH

GEAN [3] includes ILS [4], DVOR, DME, and NDB. Sari [5] explored the handling methods of RSCU monitor errors. The study focused on the DVOR system at an international airport in Indonesia, analyzing the remote control unit display errors that occurred in practical applications and their impact on the airport's navigation system. The research indicated that RSCU monitor errors are typically caused by several factors: equipment malfunctions, unstable power supply, signal interference, and improper operation. The research findings by Sari [5] provide a valuable contrast and complement to the study by Sanjaya et al. [3]. Both studies focus on technical faults within the DVOR system. Abrar et al. [6] primarily explore how to utilize monitoring tools (CCA tools) to perform corrective maintenance on DVOR, analyzing the impact of different maintenance strategies on equipment stability. Shcherbinin et al. [7] employed numerical simulation methods to analyze the performance of DME, considering the impact of different environmental conditions (such as climate, terrain interference, equipment failures, etc.) on measurement accuracy. The proposed modelling approach helps engineers to assess the robustness of the system and make the necessary adjustments to ensure the stability of the navigation system. Sandmann and Garbe [8] proposed an innovative method that uses three-dimensional visualization technology to intuitively present the variations in azimuth error texture within the DVOR system under different interference scenarios. The study combined numerical computation with simulation modeling to reveal the impact of various sources of interference (such as multipath effects, antenna asymmetry, etc.) on azimuth errors. Khadonova et al. [9] proposed an innovative "Digital Smart Airport" system based on advanced navigation and information technologies. This system integrates modern navigation techniques, satellite positioning systems (such as GNSS and GLONASS), as well as sensors and Internet of Things (IoT) technologies to achieve efficient management and monitoring of airports and aircraft. The study explores the integration of intelligent navigation systems with other intelligent systems. Many scholars have studied GEAN in terms of stability and accuracy [10], [11], [12], but little research has addressed the question of how to efficiently monitor on running status of DVOR, and most of these studies have mostly CNS Personnel how to learn the DVOR.

B. CURRENT STATUS OF DIGITAL TWIN RESEARCH

DT are increasingly being applied in industrial production [13]. It has been extensively researched by numerous

scholars. Xiong and Wang [14] conducted a comprehensive review of the applications of DT technology in the aviation industry, emphasising its significant potential to improve operational efficiency and equipment maintenance. In particular, the integration of DT into GEAN was noted for its ability to make equipment maintenance and fault diagnosis more intelligent and accurate. In addition, Liu et al. [15] suggested that the application of DT extends beyond the management of individual devices; it can be combined with Global Navigation Satellite Systems (GNSS) to perform holistic analysis and optimisation of flight navigation equipment performance. This integrated approach provides a more comprehensive health monitoring and early warning system for GEAN. Javaid et al. describes a DT as a software model of a physical thing [16]. DT support real-time monitoring of physical things [17], [18], [19]. It is driven by data and analytics [20], [21]. DT utilize data, machine learning (ML), and the Internet of Things (IoT) to enhance the efficiency of systems and enterprises [22], [23], [24]. Due to the successful application of DT in data monitoring, employing DT for remote management of DVOR has become feasible. Based on these studies, the DT system of DVOR enables preventive maintenance of equipment. This approach increases equipment reliability, reduces troubleshooting time and improves operational safety. This makes the use of DT on DVOR significant. Bachelor et al. [25] focused their research on leveraging these technologies to enhance the lifecycle management of aeronautical systems. By integrating the concepts of DT and digital threads, they proposed a novel approach to improving the design and optimization of complex aeronautical systems. This approach advanced the application of Model-Based Design (MBD) in the aerospace sector, offering new perspectives and technological solutions for the design and optimization of future aeronautical systems. Similarly, Marino et al. [26] investigated product lifecycle management and highlighted the critical role of requirements definition in the development of DT platforms.

C. CURRENT STATUS OF VIRTUAL REALITY RESEARCH

VR technology has shown tremendous potential in operational training [27], [28], troubleshooting and simulation testing of GEAN by creating immersive virtual environments. Through these virtual environments, operators can simulate operations without the need for real equipment, enabling them to test and troubleshoot, thereby improving training efficiency and operational safety [29]. Garbarino et al. [30] proposed a VR-based platform for aerospace functional testing. This platform effectively enhances operational safety and efficiency by allowing operators to simulate various flight scenarios in a safe, risk-free environment until they have mastered key operational skills. Additionally, Ghita et al. [31] discussed the application of VR in the operational rehearsal and maintenance training of aviation equipment, particularly in enhancing the skills of pilots and maintenance personnel. VR technology helps operators improve their ability to handle

complex situations by simulating various equipment failures and flight navigation scenarios. Through VR, maintenance personnel can train in equipment operation and fault handling without relying on actual equipment, significantly reducing training costs and improving operators' understanding of the equipment.

In education, an increasing number of VR technologies are being utilized. In recent years, scholars have made discoveries in research on the application of VR technology in education. Fitria argues that immersive technologies enhance students' learning experiences in unprecedented ways, making the integration of VR technology into education a topic worthy of schools' attention [32]. After conducting a survey on VR systems for computer assembly, Bramantya et al. concludes that the use of VR media can enhance learning outcomes [33]. Ahmad et al. investigated the assistance provided by augmented reality technology in locating acupuncture points during cupping therapy [34]. In analyzing student learning, Gao et al. applied Cognitive Diagnosis to assess students' mastery levels of various knowledge concepts [35]. VR technology has made efficient learning of DVOR possible. The application of VR to DVOR enables learners of DVOR to engage deeply in the learning process, improving memory and comprehension. Training organisations are also able to save on the cost of purchasing expensive DVOR, allowing them to train people to a high standard, but at a lower cost of use.

D. CURRENT RESEARCH STATUS ON THE INTEGRATION OF DIGITAL TWINS AND VIRTUAL REALITY

The integration of DT with VR has made the operation, maintenance and training of GEAN more intelligent and interactive. By combining real-time data from DT with the interactive interface of VR, operators can monitor equipment status in real time in a virtual environment, simulate operations and troubleshoot, thereby improving operational efficiency and safety. Ghita et al. [31] proposed that the fusion of DT and VR offers innovative solutions for equipment management in the era of Aviation 4.0, providing operators with comprehensive decision support through the integration of virtualisation and real-time data. This combination is not only applicable to operational training, but also plays a crucial role in equipment monitoring and troubleshooting. Vairagade et al. [36] explored the combined application of DT and VR technology in nuclear power plants, which also has potential for the maintenance of GEAN. Bugaj [37] propose a comprehensive framework for the application of various cutting-edge technologies in virtual economies, especially in the context of the metaverse. Bugaj's study advances the use of DT, Internet of Things, big data computing, and image recognition technologies within virtual economies, offering both theoretical and practical support for future technological innovations. By integrating VR with DT, an immersive training environment can be provided to help operators respond more quickly and accurately to equipment failures.

E. SUMMARY

DT technology is a key tool in modern aviation and has significant application prospects in GEAN's operational monitoring, fault prediction and maintenance optimisation [14]. In particular, in the performance analysis of GEAN such as DVOR, DT enables real-time monitoring of equipment status through virtual modelling, facilitating early detection and prediction of potential failures. The integration of DT and VR technologies within GEAN is beginning to mature and shows significant potential. DT provides real-time monitoring, condition analysis and fault prediction for equipment, while VR enhances operator skills and emergency response capabilities through immersive interactive environments. Together, these technologies not only improve the operational efficiency of GEAN, but also offer new opportunities for intelligent equipment management and training. As related technologies evolve, the deep integration of DT and VR will become a major trend in the aviation industry.

III. OVERALL SYSTEM DESIGN

This paper provides a novel DVOR data acquisition algorithm for realizing a DT system of DVOR to handle the issue (a). And proposed a novel method for constructing DVOR fault logic rules to realize a virtual simulation teaching system for DVOR, thereby addressing the issue (b). Given DT's successful application in the industry field, in order to solve the issue (a), this paper applies DT technology. DT contains three essential concepts: physical entity, virtual entity, and data mapping. In the beginning, the physical entity is the DVOR under study. How to realize the DT data mapping is a key determinant of the DT system. DVOR running data can be used to acquire by serial port. Therefore, It is obvious that serial port communication resorts to a data collector. To satisfy the C/S framework, Raspberry Pi 4B is desirable hardware as a data collector. Then, the data acquisition algorithm is necessary to be realized. It's not enough to have the data acquisition algorithm, there is also a need three-dimensional simulation environment. Several techniques have been developed to realize the three-dimensional simulation environment. Furthermore, Unity is one of the most popular techniques applied in the DT field [38], [39]. Therefore, this paper chooses Unity3D to construct a VR environment. The DT entities are constructed by Blender which is a lightweight modeling tool. Given that VR technology is used in the teaching course and the cost is cheap [40], [41], [42], this paper chooses VR technology to solve the issue (b). Finally, only to implement the system is not enough to justify its usefulness and worth. So, in Section VII, questionnaire assessments of the system were collected from 40 volunteers who participated in the system trial operation. It also includes experiments and discussions on the scalability of the system architecture and the minimum data requirements for the system.

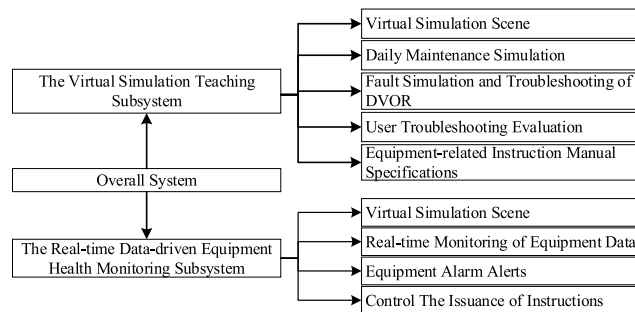


FIGURE 1. System functional structure diagram.

This study provided the DT system consists of divided into two main subsystems: The Real-Time Data-Driven Equipment Health Monitoring Subsystem (RTDEHMS) and The Virtual Simulation Teaching Subsystem (VSTS). The overall functional structure of the system is shown in Fig. 1.

RTDEHMS is functionally divided into five aspects: virtual simulation scene, real-time equipment data monitoring, equipment alarm alerts, and control of instruction issuance. This subsystem enables remote management of the DVOR, providing both monitoring of its running information and rapid warning for the equipment manager. RTDEHMS successfully resolved issue (a).

VSTS can be classified depending on function into virtual simulation scene, daily maintenance simulation, fault simulation and troubleshooting of DVOR, user troubleshooting evaluation, and equipment-related instruction manual specifications. Resorts to those functions the DVOR learners are able to immediately familiar with the basic operation of DVOR. Furthermore, the fault simulation and troubleshooting function of this subsystem may have contributed to the increase in CNS Personnel troubleshooting capabilities. VSTS can provide a solution for issue (b).

Referring to the five-dimensional model of DT based on Schroeder's [43] and DT design framework of Fei Tao's [44], [45], the overall system architecture is shown in Fig. 2.

A. PHYSICAL LAYER

The physical layer is an essential part of the DT system. In this paper, the physical entity of DT is the DVOR (type: VRB-52D) which is produced in Italy. It is not allowed to modify the DVOR, but the DVOR running information can be acquired by serial port. On the one hand, resorting to serial port communication, the DVOR transmits the hexadecimal string to external. These strings contain the DVOR real-time running information. These strings are not the same at different times. Therefore, this information constitutes dynamic data which is shown in Fig. 2. The static data contain the equipment appearance, antenna size, and number of antennas. It will be used for the construction of a virtual environment.

The reason for choosing the VRB-52D as the research target in this paper is that the VRB-52D is more complex than

other GEAN. From the perspective of building a DT, it has a very large number of buttons, knobs, indicators, test ports, and other components, which poses a significant challenge for the construction of a DT. Although this paper has used the VRB-52D equipment as the physical layer of the DT, the system architecture is not only applicable to different models such as the DVOR4000 equipment, but also to different types of GEAN such as, ILS, NDB, and MB. given the current data acquisition methods of GEAN, others can be carried out.

The minimum system requirements and relevant application scenarios for this approach will be presented in subsection VII-B.

B. DATA MAPPING LAYER

This paper uses Raspberry Pi 4B as the data collector. The RPi 4B is an excellent Single Board Computer, performing well in all tests and significantly outperforming other models of Raspberry Pi [46]. From a cost and performance standpoint, the 4B is very suitable hardware for use as a data collector. The data collector is responsible for acquiring data from physical entities, parsing, encapsulation, and driving DT entities with dynamic data. On the one hand, data is stored and forwarded through an embedded MySQL database and a Java server within a Client/Server architecture. On the other hand, the data collector can receive operation commands that are formed by user operation actions in the VR environment. Java offers cross-platformity, high performance and scalability, rich framework and library support, and security, making it a suitable programming language for developing stable, scalable, and secure server applications. Choose MySQL for ease of use and flexibility. MySQL offers a wide range of security features, including encryption, access control, and rights management. It supports data encryption, SSL connections, and enhanced security through granular control of accounts and roles. MySQL has many third-party tools and plug-ins to enhance its functionality, such as backup tools, performance analysis tools, and more!

C. VIRTUAL ENTITY LAYER

The overall system architecture consists of RTDEHMS and VSTS. RTDEHMS's virtual entities require only DVOR, ground architecture and antennas.

VSTS is as close as possible to the real environment. In their daily maintenance work, CNS Personnels often use oscilloscopes, multimeters, and power meters to troubleshoot equipment. In order to meet the needs of fault simulation and diagnosis, it is necessary to provide a set of virtual test tools. In this paper, virtual tools are simulated by presetting information in advance. Take the virtual oscilloscope as an example, first of all, through the modelling to build the appearance model of the virtual oscilloscope, through the preset information to load the relevant test waveforms, used to provide feedback on the user's test operation display. The virtual oscilloscope in this paper is not an element in the DT

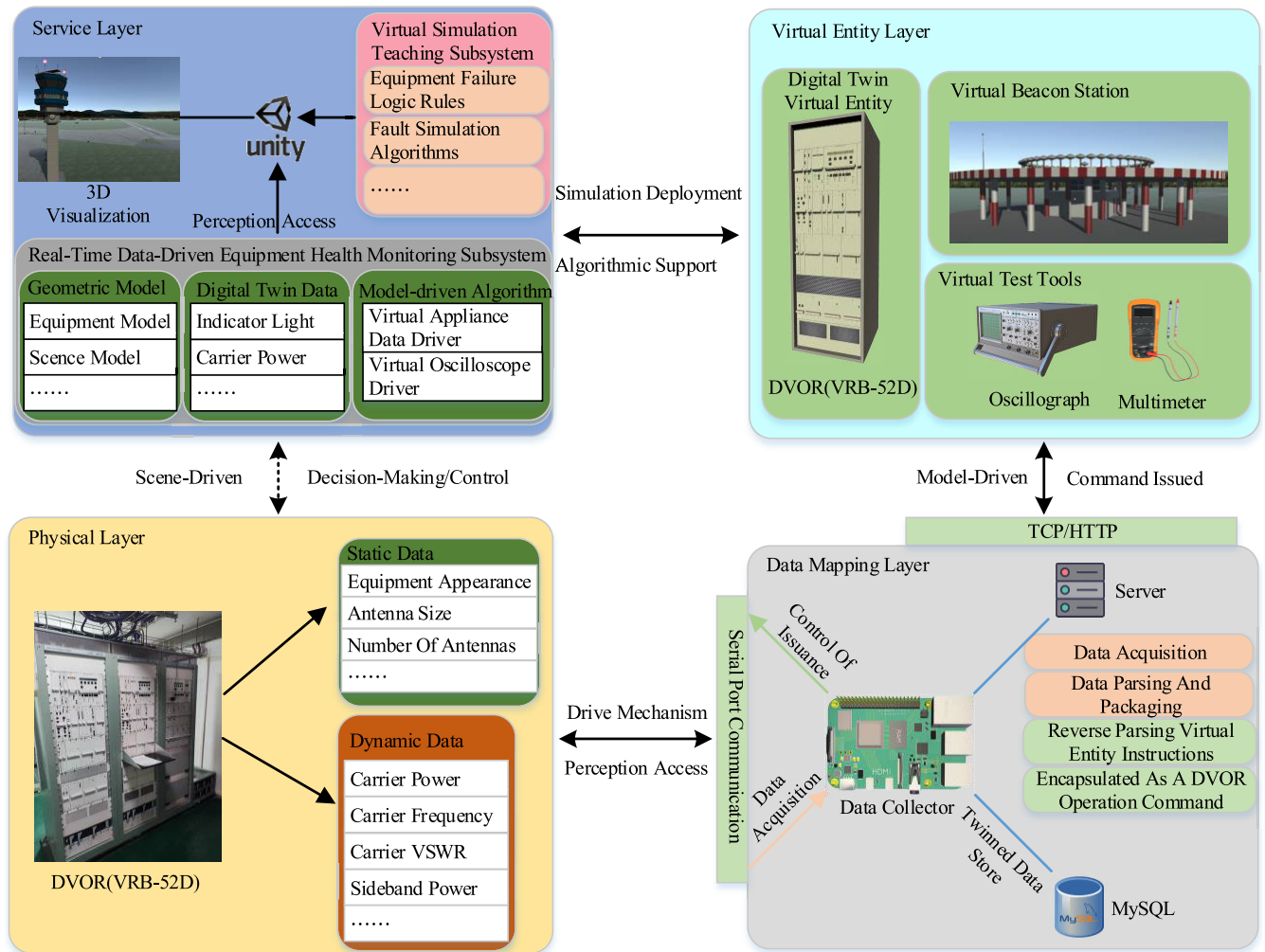


FIGURE 2. Overall system architecture diagram.

system, it is only used in VSTS, which is used to improve the process of simulating troubleshooting.

D. SERVICE LAYER

The service layer of the system architecture includes RTDEHMS and VSTS. The virtual scenes of GEAN are visualized in 3D using Unity3D.

To implement RTDEHMS, the DT service layer adopts a model-driven approach. It receives real-time data from physical GEAN (such as signal status, operational parameters, and fault information) and processes it into a format suitable for display in Unity. The operational status, performance metrics, and environmental data of the physical equipment are dynamically presented through an intuitive visual interface. Real-time fault conditions are also displayed, enabling interactive engagement between users and the virtual equipment. In Section IV, RTDEHMS will be analyzed and discussed in detail.

The service layer of VSTS integrates equipment fault logic rules with fault simulation algorithms, while also

incorporating a user operation evaluation algorithm. This creates a robust fault simulation capability that effectively supports fault diagnosis and troubleshooting tasks. Additionally, VSTS provides users with comprehensive tools focused primarily on DVOR, including standards, documentation, and query functionalities, enabling convenient access to relevant guidelines. The detailed content of VSTS is discussed further in Section VI.

In this section, it has been explained that overall system architecture and essential consists. The following section moves on to consider the RTDEHMS realization method.

IV. THE REAL-TIME DATA-DRIVEN EQUIPMENT HEALTH MONITORING SUBSYSTEM

RTDEHMS is divided into three main components, as shown in Fig.3. It comprises a physical entity component, a virtual entity component, and a bidirectional data mapping component. The physical entity module and the bidirectional mapping module communicate with each other using serial port communication. The bidirectional mapping module

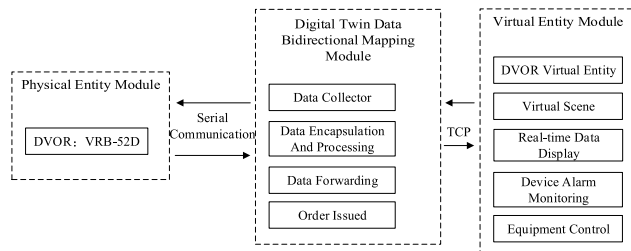


FIGURE 3. Bidirectional data mapping module.

communicates with the virtual entity module via TCP/IP communication. The most critical data mapping methods and the data transmission validity validation method will be discussed in detail in subsubsection IV-C1 and subsubsection IV-C2.

A. PHYSICAL ENTITY MODULE

This study primarily focuses on constructing a DT system using the DVOR model VRB-52D as an example. Therefore, DVOR is the physical entity module within the system architecture. DVOR consists of several modular components, each with a different function. Typically, the occurrence of a fault generally originates from one of the components. For example, if the power supply module fails, the CNS Personnel only needs to carry out the replacement and maintenance of the power supply module. By exploring the application of DT technology in specific types of equipment, the construction of a DT system for DVOR allows for the assessment of the applicability of DT technology to other GEAN. This is because the data acquisition methods for these types of GEAN are similar, typically involving serial port communication.

B. VIRTUAL ENTITY MODULE

The virtual entity module is an indispensable component of DT. Fig. 4 illustrates the process of constructing VR scenes. The model of the virtual entity can be constructed using 3D modeling software such as 3Dmax, Maya, or Blender. Alternatively, it can be reconstructed using 3D scanning technology with Slam algorithms [47], [48], or through the Neural Radiance Fields (NeRF) algorithm, which utilizes photos for model reconstruction [49], [50], [51].

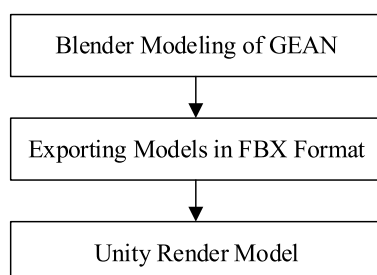


FIGURE 4. Process of constructing VR scenes.

However, mere modeling is insufficient; it is also necessary to build a three-dimensional simulation environment for VR. This environment can be created using technologies like Unity3D, Unreal, OpenGL, WebGL, etc. This paper describes the use of Blender for constructing virtual equipment which includes Alford Antennas, Monitoring Antennas, Buildings on the ground, and DVOR. Additionally, Unity 3D is employed for the construction of the DT virtual entity part. Fig. 5 depicts scene rendering for the DVOR environment work rendering.

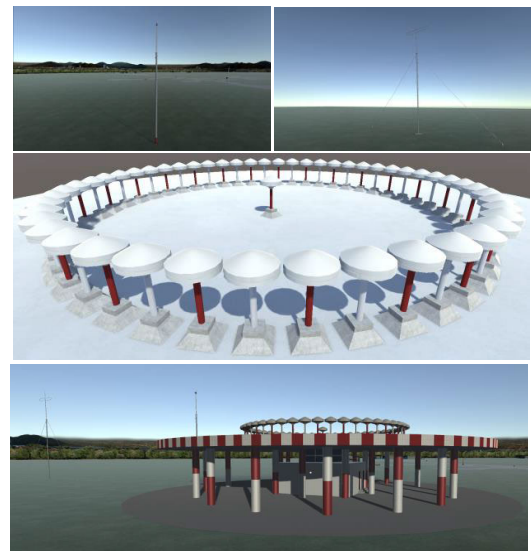


FIGURE 5. DVOR station simulation system.

Despite environment rendering being essential, creating a highly realistic model of the DVOR is even more crucial for achieving high-fidelity simulations. Fig. 6 illustrates a comparison between the physical entity's front panel and the virtual entity's front panel.

The ability to control virtual entities in Unity3D comes from the use of C# scripts. Some of the widgets in the virtual entities can be manipulated by the user (e.g. buttons, knobs, indicators, screens, etc.). The user's actions on the virtual entity will act on the physical entity. Similarly, operations on the physical entity will react on the virtual entity. This is what bi-directional mapping means at its most basic level. The following section explains the bidirectional mapping of DT data.

C. DT DATA BIDIRECTIONAL MAPPING MODULE

1) DATA MAPPING METHODS

In order to satisfy the DT model, bi-directional mapping must be accomplished between physical and virtual entities. For RTDEHMS, the real-time synchronization provided by this module ensures timely and accurate data updates, thus improving subsystem responsiveness. The complete mapping method from physical entities to virtual entities for control is illustrated in Fig. 7.



FIGURE 6. Physical entity vs. Virtual entity.

TABLE 1. VOR VRB-52D mapping data item.

Primary	Carrier Power	Rx Carrier Level	9960 Hz clock
Secondary	Carrier Modulation	30Hz AM Level	sband DRV odd
Monitor	Carrier VSWR	30Hz FM Level	sband DRV even
CTU	Lsb Sideband Power	Subcarrier Level	Keyed Ident
Bearing	Usb Sideband Power	Ident Level	Rack No.1
30hzAM	Restart Count	Carrier Forward (AV)	Rack No.2
30hz FM	30Hz Clock	Carrier Forward (PK)	Normal
Subcarrier	9960Hz Clock	Carrier Reverse	Transfer
Antenna	Commutation Freq	Lsb Sideband Level	Shutdown
Tx fw Power	400Hz Calibration	Usb Sideband Level	Antennas to Rack
Sideband 1	+24V Supply Voltage	30Hz Sine Ref	CTU Link
Sideband 2	+5V Supply Voltage	Tx Drive Level	Maintenance Mode
Ident	+15V Supply Voltage	Tx Imbalance 1 Level	Frequency Ctrl Lsb
+15V Monitor Supply	-15V Supply Voltage	Tx Imbalance 2 Level	Frequency Ctrl Usb
-15V Monitor Supply	+5V Monitor Supply	Tx Imbalance Comb	Blending Functn Even
+2.49V Calibration	Bearing	30 Hz clock	Blending Functn Odd

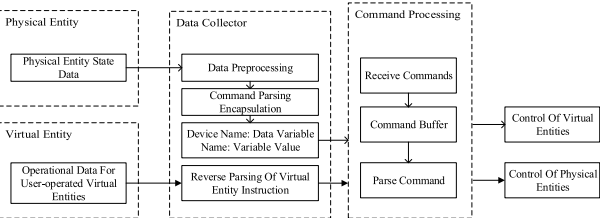


FIGURE 7. Data Mapping Process.

The running data of the DVOR is collected by the acquisition algorithm through a serial port. After a series of experimental tests, it has been confirmed that those data are in hexadecimal format. The data items collected during DVOR running are presented in Table.1.

These items include carrier power, 30Hz AM level, and commutation frequency, among others. This data includes real-time information about the operation of the equipment. By collecting real-time data, anomalies and faults can be detected promptly, providing early warnings to prevent equipment damage or system downtime, thereby increasing the reliability and safety of the equipment. In addition, these

data collection points help to monitor the operational status and performance indicators of the GEAN, assess whether the equipment is operating at optimum levels, and facilitate any necessary adjustments and optimisations.

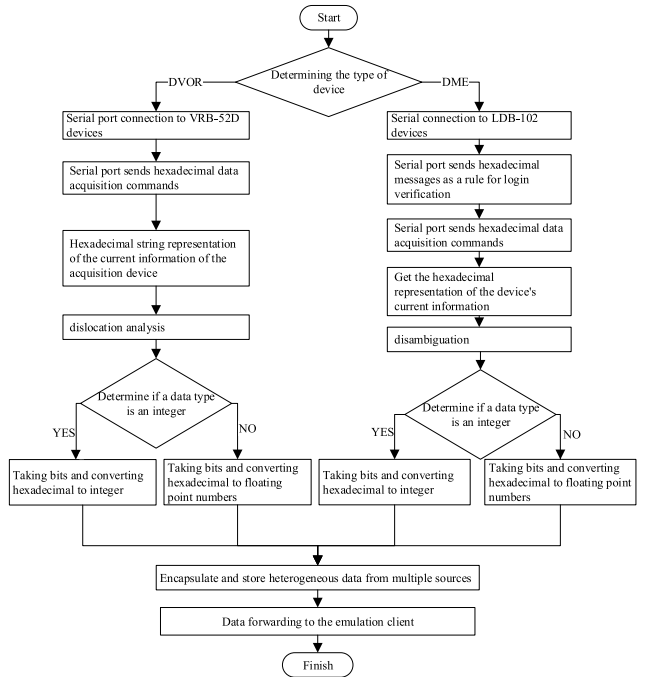


FIGURE 8. The data acquisition algorithm.

2) DATA TRANSMISSION VALIDITY VALIDATION METHOD

The data elements to be mapped have been described in the previous subsection, and data acquisition is also required to complete the data mapping. First, the data collector determines the type of GEAN. This is because the DVOR and DME share a serial port for data acquisition. The data collector then connects to the VRB-52D. Data acquisition is performed by sending a specific hexadecimal string.

The serial data acquisition parameters for GEAN, such as baud rate, stop bits, and parity, are typically determined based on the specific equipment requirements. Common configurations include 8 data bits, no parity, and 1 stop bit (commonly referred to as “8-N-1”), with baud rates of either 19200 or 115200, depending on the communication speed and application needs.

The algorithm employs a triple verification strategy. During serial port data acquisition from GEAN, if data loss occurs due to transmission issues, the algorithm records an instance of loss. If data loss occurs consecutively three times, an immediate alarm is triggered. Different types of GEAN require varying acquisition frequencies because they adhere to different serial port communication protocols, as shown in Table.2.

The information in this table can be used to understand the technical characteristics and operational requirements of different GEAN, in particular the differences in data acquisition.

TABLE 2. Acquisition model table.

Categories of GEAN	Equipment type	Acquisition model	Acquisition Methods
ILS	7033B	Proactive acquisition	Send a get command
DVOR	DVOR4000	Proactive acquisition	Connect the validation command set and send the acquisition command
DVOR	VRB-52D	Passive Acquisition	Push data to serial port every second

Active acquisition models usually require external commands to trigger data acquisition, whereas passive acquisition models automatically send data at regular intervals. This information is critical for data acquisition by GEAN.

The VRB-52D will respond with a hexadecimal string containing the current operational information. The data collector parses this hexadecimal string, with each data item represented by 16 bits in binary. Depending on the meaning of the data, it is parsed into either an integer or a floating-point number. Once the data parsing is complete, it is immediately encapsulated using the format “Device Name: Data Variable Name: Variable Value”. The data will be stored in the MySQL database embedded within the data collector. Furthermore, owing to the presence of a Java server integrated into the data collector, these data will also be distributed to the Unity3D simulation clients simultaneously. The process of the data acquisition algorithm is illustrated in Fig. 8.

In Unity3D, this study designed an instruction processing module for preprocessing. This module first identifies the category of GEAN and data variable name, then determines the virtual entity object that requires modification according to the category GEAN and data variable name. Finally, it updates the received DT data to the corresponding virtual entity.

V. GEAN DATA SECURITY

The system stability and safety of the GEAN is crucial to flight safety. As a derivative of the DT monitoring system serving the GEAN, the acquisition of real-time operational data of the equipment, transmission, stability of data transmission, delay of serial data acquisition, data integrity, etc., are unavoidable.

A. LOCAL AREA NETWORK TRESPASSING RISK CONTROL

It is understood that almost all GEAN in China do not have access to the Internet, but rather use a local area network (LAN) to ensure the security of information, and that GEAN are commonly equipped with the ability to operate the equipment remotely. Although the LAN is not connected to the Internet, it can still be compromised by malicious programs, wireless signal attacks and unauthorised access to the equipment. If this LAN is accessed by an unauthorised intruder, the equipment can be illegally switched off and the settings illegally changed, creating an uncontrollable security risk to air travel. It can be protected in a number of ways.

1) PHYSICAL ISOLATION AND ACCESS CONTROL

Ensure that the LAN is completely physically isolated from external networks and strictly limit the number of access equipments on the LAN.

2) NETWORK ACCESS CONTROL (NAC)

Deploy an access control mechanism based on MAC address and equipments authentication to ensure that only authorised equipments can access the LAN [52].

3) SEGMENTED NETWORK ARCHITECTURE

Use VLAN [53] or subnetting to isolate different functional modules and limit the impact on the whole network after an intrusion.

B. INTRUSION DETECTION AND ANOMALY PROTECTION

1) INTRUSION DETECTION SYSTEM (IDS)

Deploy an IDS [54] or next-generation firewall (NGFW) to monitor network traffic in real time and identify suspicious access behaviour.

2) BRUTE FORCE CRACKING PREVENTION

Avoid brute-force cracking attacks [55] on equipments or networks by limiting the number of login attempts, enforcing complex passwords, and changing passwords regularly.

3) PENETRATION TESTING

Conduct regular vulnerability scanning and penetration testing of the LAN to identify potential security risks [56]. Use traffic auditing tools to detect abnormal access patterns.

C. SECURITY IMPROVEMENTS IN DIGITAL TWIN SYSTEMS

The DT technology deeply integrates the physical system and the virtual model, and its safety is directly related to the reliable operation of GEAN.

1) TWO-WAY ISOLATION MECHANISM FOR DATA SECURITY

Prevent the state of the physical equipment from being tampered with through the loopholes of the DT [57]. Restrict the DT system to read-only access to the sensitive data of the equipment.

2) REDUNDANCY MECHANISM

Design redundant data acquisition channels for the DT system to prevent monitoring failure due to the failure of a single data source. Establish a mechanism to back up navigation system data in case of unexpected storage failures or attacks.

3) ZERO TRUST ARCHITECTURE

Adopting the zero-trust concept [58], authentication, equipment verification and behaviour verification are performed for every access request, even if the request comes from within the LAN.

4) BEHAVIOURAL DYNAMIC AUTHORISATION

Monitor the operational behaviour of equipments and users to establish a behavioural baseline. Automatically block and alert on abnormal operations [59].

5) AI-BASED ANOMALY DETECTION

Machine learning models are used to identify normal operating states and data transfer patterns of equipments, providing real-time warnings of potential security threats such as network attacks or abnormal equipment operation, and enhancing security by analysing abnormal network traffic such as oversized packets or frequent connection requests [60].

This section has examined the implementation of safer and more effective risk control measures within RTDEHMS of GEAN. The aim of this section is to improve the overall safety of the system.

VI. VIRTUAL SIMULATION TEACHING SUBSYSTEM

In the preceding section, RTDEHMS has been discussed. In this section, the focus will shift to VSTS, discussing the methodology for constructing logical rules for GEAN anomalies and establishing algorithms for evaluating user actions. The VSTS is intended to address the issue (b) outlined in the Introduction section. Special attention should be paid to the fact that VSTS is a purely virtual simulation system without any DT technology!

A. VSTS'S FAILURE SIMULATION CONSTRUCTION METHODOLOGY

In order to support the simulation of failures in the DVOR and to ensure that the virtual models in VR scenes exhibit operational effects consistent with real equipment, a methodology is proposed for the construction of the failure simulation of VSTS. The anomaly logic rules for DVOR encompass two main aspects:

- **Equipment Fault Logic:** Causality between DVOR faults, their interrelation, and the fault components corresponding to the fault phenomena.
- **Equipment Fault Factors:** Abnormal factors of faults, namely the causes of faults.

These two aspects will be elaborated in detail in the following subsubsection VI-A1 and subsubsection VI-A2.

1) EQUIPMENT FAULT LOGIC CONSTRUCTION METHOD

Equipment fault logic refers to the objective laws that lead to equipment faults in practical scenarios. It includes the causes of faults, the causal relationships behind faults and the relationships between different faults. In order to construct the equipment fault logic, this paper conducts analysis on various aspects of the DVOR, including: (e.g., DVOR principles, DVOR fault feedback indications, DVOR fault inspection checklists, etc.). Fig. 9 outlines the elements that need to be analyzed for the equipment fault logic.

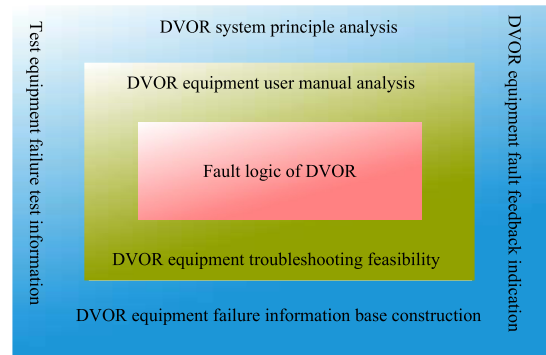


FIGURE 9. Methods for constructing equipment fault logic rules.

After completing the comprehensive analysis of various aspects of DVOR, it is necessary to abstractly extract and store the objective laws obtained from the analysis. Given the need to simulate various fault scenarios for DVOR, this paper proposes a data structure representation method for fault logic. This data structure is illustrated in Fig. 10. The fault logic of DVOR is represented by a directed graph constructed with DVOR faulty logical nodes. The nodes in the directed graph represent specific faults, while the edges represent relationships between the faults.

The data structure facilitates an easier description of fault information, which is beneficial for programmers in developing fault simulation functions. By switching fault scenario information, it enables the tracing back of faults and tracking of fault handling methods. Loading the UCSD of any node allows tracing back to the state of that fault handling. Additionally, this data structure effectively compares user operations with FHSP, providing evaluation and scoring for user actions.

The directed graph not only represents the connections and causal relationships between faults but also stores relevant information about the faults. The fault information of the DVOR is encapsulated within nodes, and the data structure of each node is visualized and illustrated in Fig. 10. This data structure contains several elements of fault simulation. In the structure of node e , the stored fault information includes (Fault Association(FA), Fault Name (FN), Fault Diagnosis (FD), Fault Component (FC), Fault Standard Handling Procedures (FSHP), Unity Corresponding Scene Data (UCSD)). Those information stored in these nodes will be utilized in the implementation of fault simulation and troubleshooting methods discussed in subsection VI-B.

In Fig. 10, the DVOR shutdown fault logic structure is presented as an example to illustrate how to establish a DVOR Fault Logic. Table 3 provides specific meanings for the nodes. It can help to understand how to represent DVOR fault logic.

Among these, $\{e(0, i), i = 0, 1\}$ represents faults that may lead to $e(0)$. The potential causes of fault $\{e(0, i), i = 0, 1\}$ are represented by $e(0, i, j)$, where the maximum value of j

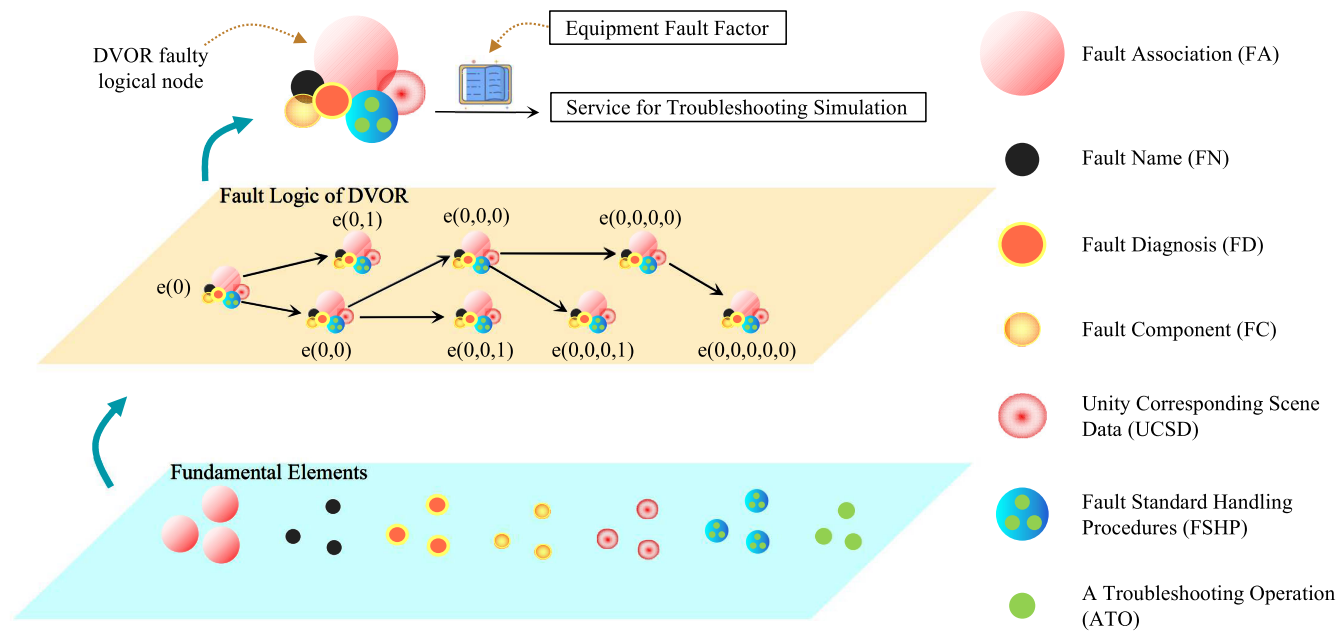


FIGURE 10. The DVOR fault logic rules.

TABLE 3. DVOR VRB-52D mapping data item.

Node	Meaning
e(0)	VRB-52D DVOR failure shutdown station
e(0,0)	Failure in the transmitter and monitor
e(0,1)	Power supply failure in the control unit
e(0,0,0)	Fault in the carrier modulation signal module
e(0,0,1)	CPA component failure
e(0,0,0,0)	Fault in components related to carrier modulation signals
e(0,0,0,1)	Monitoring channel failure or external field environment
e(0,0,0,0,0)	Fault in components related to carrier modulation signals

corresponds to one less than the possible number of causes, and the minimum value of j is 0. For the subscript notation of e , the format $(x_1, x_2, x_3, x_4, \dots, x_i)$ is used. If a fault occurs with the subscript $(x_1, x_2, x_3, x_4, \dots, x_i)$, it suggests that a fault corresponding to $(x_1, x_2, x_3, x_4, \dots, x_{i-1})$ will also manifest. This allows for fault localization to be carried out layer by layer through progressive indexing. These subscript notations are stored in the fault association elements of the fault logic nodes.

2) EQUIPMENT FAULT FACTOR INTEGRATION METHOD

During the DVOR fault simulation process, it is essential not only to represent fault logic rules consistent with real equipments but also to provide users with scenario simulations of equipment failures caused by unforeseen circumstances. The study compiles several common scenarios, such as interruption of utility power, antenna anomalies, and voltage instability. The occurrence of these unforeseen events may lead to component failures in DVOR. This enables the association of emergency situations with the fault nodes depicted in Fig. 10. This enables mutual fusion between

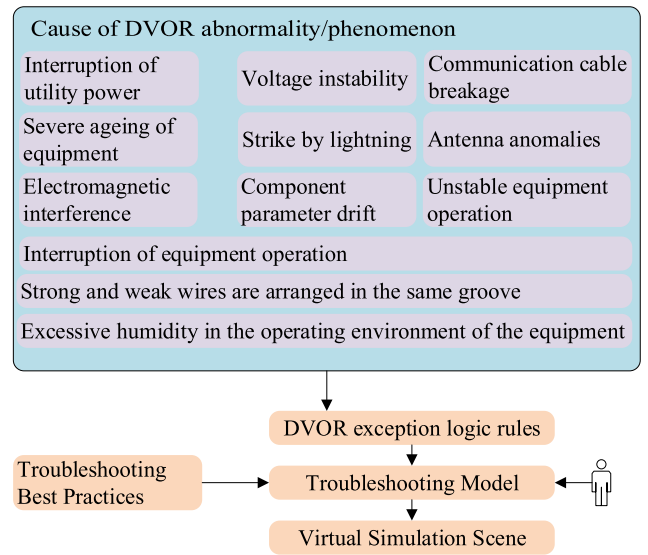


FIGURE 11. Multiple heterogeneous logic rule fusion method.

equipment fault factors and fault logic rules. This makes it possible to provide users with simulated equipment failure scenarios caused by unexpected situations. The integration of equipment fault factors with equipment anomaly logic is illustrated in Fig. 11. The reasons behind the DVOR anomalies are categorized based on the information we were able to collect.

In this subsection, the construction of fault logic rules and the integration of the integration of equipment fault factors. Details regarding the simulation of DVOR faults are in the next subsection.

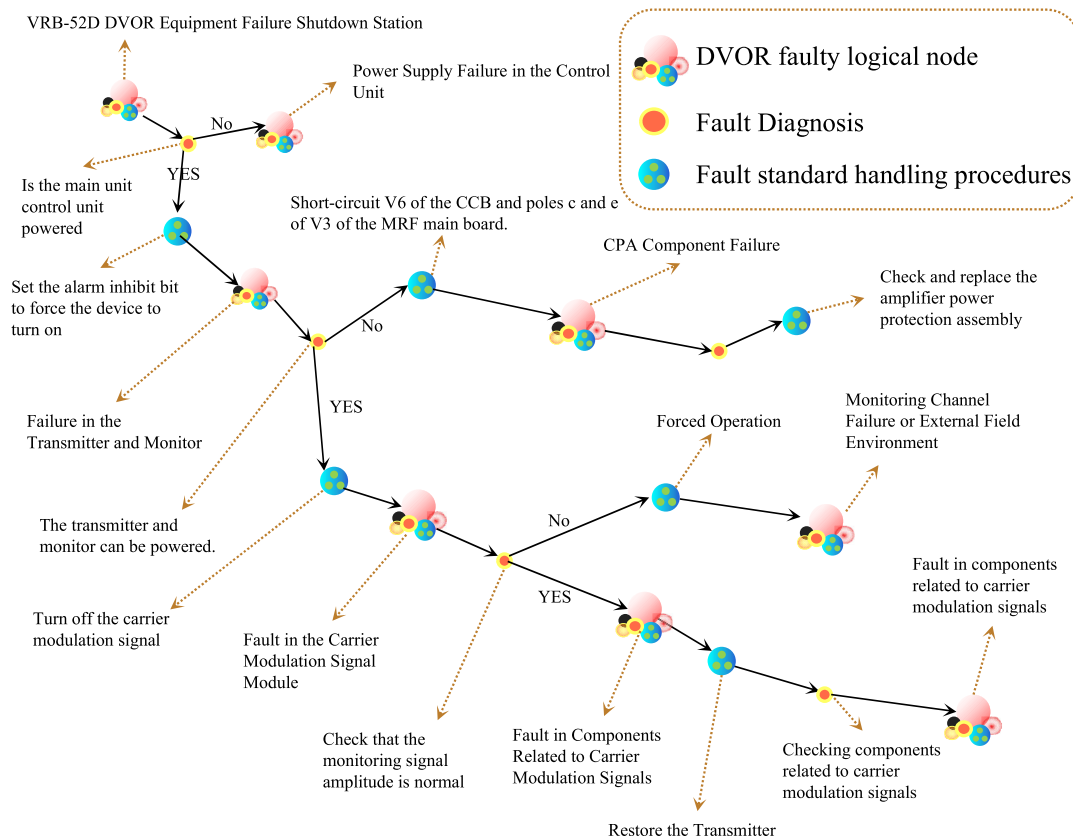


FIGURE 12. Example of troubleshooting process for a shutdown station.

B. FAULT DIAGNOSIS EVALUATION ALGORITHM FOR USERS IN UNITY

In the preceding segment, the paper has elucidated the process of constructing equipment anomaly logic rules.

The following segment will provide a detailed explanation of DVOR fault diagnosis simulation in Unity, as well as the evaluation method for troubleshooting faults by users.

1) DVOR FAULT DIAGNOSIS SIMULATION IN UNITY

In the preceding subsection, the data structure for storing fault information has been described. To implement DVOR fault diagnosis simulation in Unity, different scenarios corresponding to various faults need to be implemented. To showcase these scenarios, it's necessary to pre-store scene state information. This information includes the color of indicator lights, the status of switches, the data displayed on screens, and the position of knobs. With this information, the visual cues presented to users regarding the equipment's status during a specific fault can be effectively conveyed.

The UCSD in the fault logic nodes stores this information. This indicates that each fault node corresponds to a scenario. The scene information corresponding to fault nodes in the DVOR fault logic rules is stored as a class object. In Unity, storing scene states can be accomplished using JavaScript Object Notation (JSON) format, which offers advantages such as data persistence, easy storage, and loading. The FD in

the fault logic nodes stores the standard diagnostic procedures users employ for the current fault. In the fault logic nodes, FSHP stores the correct procedures users should follow for the current state of the DVOR. In the fault logic of DVOR, there is a sequence between fault diagnosis and fault handling operations, depending on the specific circumstances. These standard fault diagnosis or equipment operation steps serve as benchmarks for evaluating the user's fault troubleshooting process. Taking a DVOR station shutdown fault as an example, how DVOR fault logic and fault nodes are utilized in fault troubleshooting has been illustrated in Fig. 12.

In practice, the fault diagnosis process of DVOR often involves using diagnostic tools to measure data. By providing virtual oscilloscopes, multimeters, network analyzers, and power meters, users can conduct signal testing, fault troubleshooting, and analysis of virtual DVOR in the virtual world.

As CNS Personnel are required to carry out regular maintenance on the DVOR, the subsystem includes a daily maintenance simulation capability to provide a more efficient and immersive learning experience for users. This functionality allows users to operate normally functioning equipment as part of their routine maintenance training. During maintenance procedures, it is common to test the subcarrier modulation degree of DVOR. Using this test as an

example, Fig. 13 illustrates how an oscilloscope is utilized to test the subcarrier modulation degree of DVOR.

Using a virtual oscilloscope, compare the waveform with the standard waveform (as shown in Fig. 13). The standard waveform shows what the output waveform should look like if the output port is normal. If this is not the case with the virtual oscilloscope, it means that something is wrong with the 9960Hz subcarrier generation or modulation process, and appropriate troubleshooting and processing is required.

$$AM = \frac{(Amplitude_{max} - Amplitude_{min})}{(Amplitude_{max} + Amplitude_{min})} \times 100\% \quad (1)$$

The maximum and minimum amplitude values are recorded in the virtual oscilloscope and then the subcarrier modulation regime is calculated using Equation (1), if this modulation regime is not within the standard range then it indicates that there is a problem with the modulation waveform process.

2) THE EVALUATION METHOD FOR TROUBLESHOOTING FAULTS BY USERS

The assessment of user troubleshooting is performed based on the troubleshooting standard operating steps mentioned earlier. These standard steps are stored in the JSON format within the fault nodes. In a user's DVOR fault troubleshooting, the evaluation method records the user's input operations. Evaluation is conducted by comparing the differences between the user's input operations and the standard processing procedures. A score is given based on the completion of the user's fault troubleshooting for a single fault. In the case of a DVOR (VRB-52D) shutdown fault, The method for assessing users' fault troubleshooting capabilities is depicted in Fig. 14.

In this figure, E represents a specific fault node while S represents the score. The fault simulation algorithm loads an E from the DVOR fault logic into the Unity scene. the DT entities show the correspondent fault phenomenon. The user performs diagnostics and operations within the virtual scene. The evaluation algorithm analyses the user's diagnostic and handling steps based on the FD and FSHP within the fault nodes and gives S .

As shown in Algorithm 1, the input parameter for the algorithm is the existing fault logic rule set E . The function returns a score S on a 100-point scale. The variable x is used to record the number of operations performed. The variable n represents the minimum number of nodes traversed to complete the investigation of the fault's anomalous logic rule, indicating the minimal number of operations required. Initially, node is set to read the first anomalous logic node from the fault logic rule. In subsequent iterations, the next anomalous logic node is read. The *LoadScenes* function, via the *GameManager* global object in Unity, loads the scene data stored in the node object. A while loop is employed to iterate through the anomalous logic rule, calculating the number of operations, denoted as x , required for the user to perform the normal operations. Finally, the score S is

Algorithm 1 DVOR Fault Simulation and Troubleshooting User Operation Evaluation Algorithm

Require: E {Input: A Equipment Failure Logic Rule}

Ensure: S {Output: User score}

```

1:  $x \leftarrow 0$ 
2:  $n \leftarrow 0$ 
3:  $node \leftarrow E(n)$ 
4:  $T \leftarrow NULL$ 
5: while  $node \neq NULL$  do
6:    $n = n + 1$ 
7:    $GameManager.LoadScenes(node)$ 
8:    $currentOperation \leftarrow WaitingUserOperate()$ 
9:    $T.AddOne \leftarrow currentOperation$ 
10:   $t \leftarrow T(x)$ 
11:  while  $node.ATO \neq t$  do
12:     $currentOperation \leftarrow WaitingUserOperate()$ 
13:     $T.AddOne \leftarrow currentOperation$ 
14:     $x \leftarrow x + 1$ 
15:     $node \leftarrow node.child$ 
16:  end while
17: end while
18:  $\delta \leftarrow \frac{n \times (n-1)}{2} + 1$ 
19:  $S \leftarrow 100 \times e^{-\frac{(x-n)^2}{32 \times \delta}}$ 

```

computed using the Equation (2).

$$\delta = \frac{n \times (n-1)}{2} + 1$$

$$S = 100 \times e^{-\frac{(x-n)^2}{32 \times \delta}} \quad (2)$$

In this paper, a method is provided for the calculation of the score function $S(x, n)$, with independent variables x and n . The image of the $S(x, n)$ is shown in Fig. 15, S is the score, ranging between 0 and 100. The curves of different colors represent the $S(x, n)$ plots under different values of n . This formula uses a Gaussian function to calculate the score, reflecting how the difference between x and n impacts the score. When x is close to n , the exponent term approaches 0, resulting in $e^0 = 1$, and the score S approaches 100. When x deviates significantly from n , the exponent term becomes negative, and the score S decreases significantly. The denominator $32 \times \delta$ acts as a tuning parameter, controlling the rate at which the score decreases. The larger the value of δ , the slower the score decreases as x deviates from n . This function evaluates how well x matches the target value n . The score S reflects the relationship where the closer x is to n , the higher the score; the farther x deviates from n , the lower the score. The variable n not only represents the number of equipment abnormal logic nodes traversed in this iteration but also signifies the optimal number of operations required for the user to complete the fault diagnosis. To cultivate high-quality CNS Personnel, this paper requires that as the complexity of the simulated fault increases (with larger n values), students should use fewer operations (with x smaller

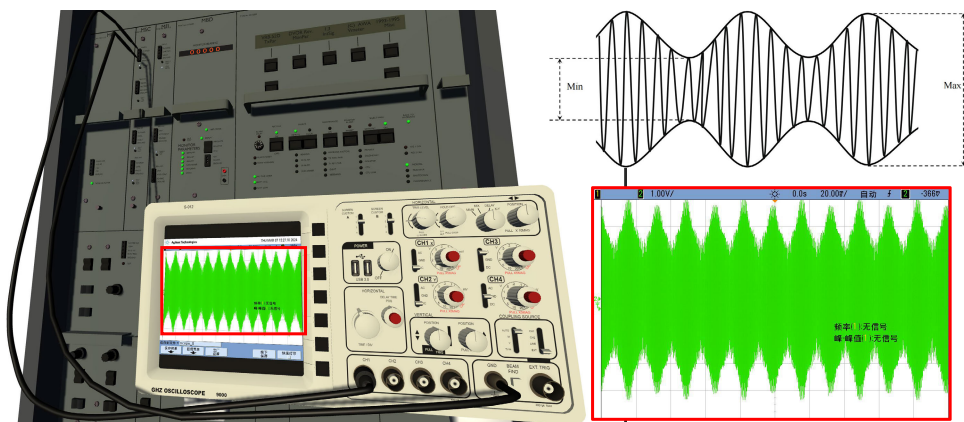


FIGURE 13. Tested on virtual DVOR’S 9960HZ FILTERED.

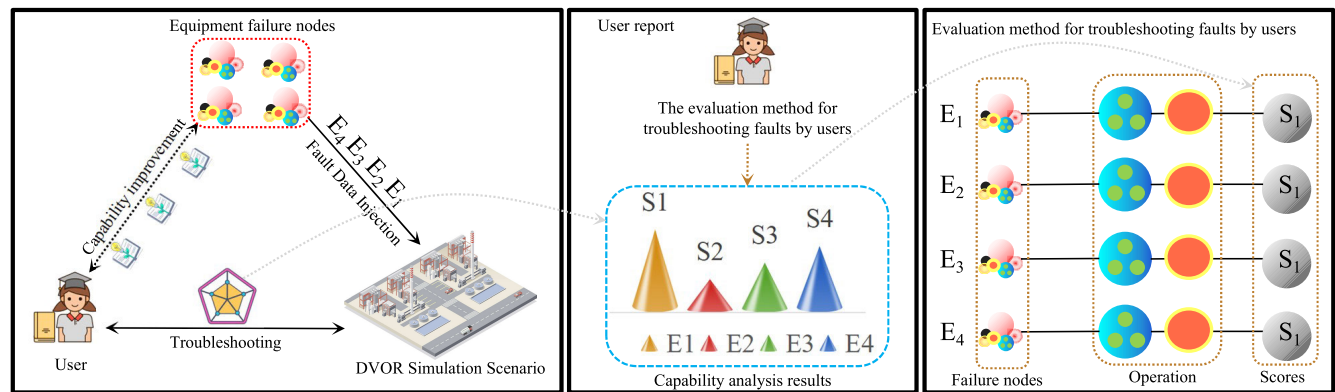


FIGURE 14. The evaluation method for troubleshooting faults by users.

values) to complete the fault diagnosis in order to achieve higher scores (with S larger values).

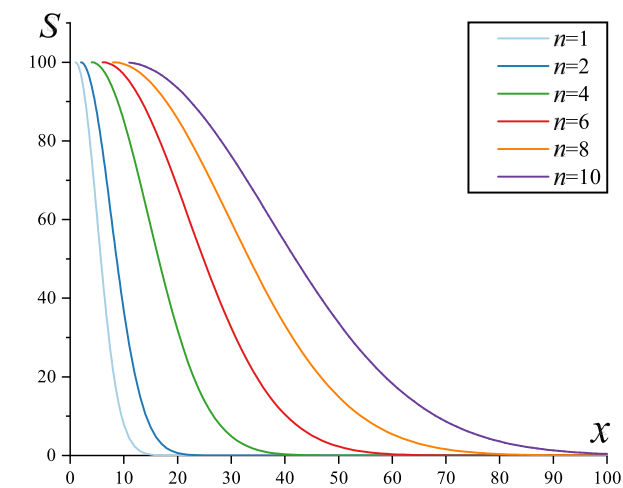


FIGURE 15. Evaluating function image.

This section presents the design and implementation of VSTS. It also details a proposed method for simulating

DVOR failures. Aspects such as data structure, three-dimensional scene construction, and evaluation of user actions are discussed.

VII. RESULTS
A. ASSESSMENT OF THE SYSTEM’S EFFECTIVENESS THROUGH SURVEY

To validate whether the DVOR RTDEHMS and VSTS addressed the issues (a) and (b) mentioned in the introduction, 40 volunteers were recruited for testing in this study. The composition of volunteers is shown in Table 4.

TABLE 4. Composition of subjects.

Personnel type	Composition of personnel	Number of persons
P1	Undeclared students in navigation	20
P2	Students who have graduated from the navigation program but have not obtained a CNS Personnel license	10
p3	Staff members who have obtained CNS Personnel licenses	10

As this system is not intended for the general public, it is aimed at personnel who wish to be, or are, employed in the

TABLE 5. Survey questions for each participant.

No.	Questions Detailed Descriptions	Responses
Q1	The extent to which using DVOR's VSTS supports and helps you	1—Strongly Disagree; 2—Disagree; 3—Neutral; 4—Agree; 5—Strongly Agree
Q2	Using DVOR's VSTS, you learn about the extent of DVOR's equipment.	1—Strongly Disagree; 2—Disagree; 3—Neutral; 4—Agree; 5—Strongly Agree
Q3	Using DVOR's VSTS, you mastered common troubleshooting.	1—Strongly Disagree; 2—Disagree; 3—Neutral; 4—Agree; 5—Strongly Agree
Q4	Using DVOR's VSTS, you learnt about common faults.	1—Strongly Disagree; 2—Disagree; 3—Neutral; 4—Agree; 5—Strongly Agree
Q5	Using DVOR's VSTS, you are able to perform routine troubleshooting and disposal.	1—Strongly Disagree; 2—Disagree; 3—Neutral; 4—Agree; 5—Strongly Agree
Q6	Using DVOR's VSTS, you are able to effectively communicate with them to discuss and complete troubleshooting.	1—Strongly Disagree; 2—Disagree; 3—Neutral; 4—Agree; 5—Strongly Agree
Q7	With DVOR's VSTS, you'd think you'd already be able to troubleshoot real DVOR.	1—Strongly Disagree; 2—Disagree; 3—Neutral; 4—Agree; 5—Strongly Agree
Q8	Using DVOR's VSTS, do you think VR teaching should be applied on other GEAN.	1—Strongly Disagree; 2—Disagree; 3—Neutral; 4—Agree; 5—Strongly Agree

TABLE 6. Results of the questionnaire survey.

Personnel type	No.	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
P1	Strongly Disagree	0	0	0	0	0	0	0	0
	Disagree	0	0	0	0	0	0	0	0
	Neutral	0	0	0	0	0	0	0	0
	Agree	11	8	5	9	7	8	6	4
	Strongly Agree	9	12	15	11	13	12	14	16
P2	Strongly Disagree	0	0	0	0	0	0	0	0
	Disagree	0	0	0	0	0	0	0	0
	Neutral	0	0	0	0	0	0	0	0
	Agree	4	6	5	3	4	7	2	9
	Strongly Agree	6	4	5	7	6	3	8	1
P3	Strongly Disagree	0	0	0	0	0	0	0	0
	Disagree	0	0	0	0	0	0	0	0
	Neutral	0	0	0	1	0	0	3	0
	Agree	3	6	5	6	5	3	3	6
	Strongly Agree	7	4	5	3	5	7	4	4

protection of GEAN. In this paper, the user population is divided into three categories: P1, P2 and P3. The P1 type can effectively evaluate people who are beginning to explore this domain. Type P2 is used to effectively evaluate the impact of this system on the capability improvement of users who have completed professional training in higher education institutions. The P3 type can evaluate the system's assistance to the personnel employed in the operation and maintenance of the navigational equipment.

These volunteers comprised 20 students majoring in GEAN, 10 graduates who had not yet obtained their licenses as CNS Personnel, and 10 CNS Personnel. The testing was conducted using computers equipped with a GTX3060 graphics card and an AMD 5700G CPU. As part of this study, volunteers were required to undergo basic software training in order to be able to use the DVOR's RTDEHMS and VSTS. Participants voluntarily use the system for two hours and then participate in questionnaires. The questionnaire, shown in Table 5, was used to evaluate the effectiveness of DVOR's

VSTS in student learning. It mainly includes the following aspects:

1. Overall evaluation (Q1): to find out how the VR system helped the students' learning as a whole.

2. Learning effectiveness (Q2-Q7): to examine the students' knowledge of the equipment, troubleshooting ability, ability to communicate with others, and whether they can apply it to the actual operation of the equipment that they have acquired through VR learning.

3. Extensibility (Q8): to explore whether the VR training could be extended to GEAN and the instructors' assessment of the students' knowledge improvement.

The purpose of the survey was to assess the effectiveness of VR instruction and to explore whether it could be applied to other navigation equipment. A Likert scale from 1 to 5 was used to measure the responses of the survey participants. If the majority of students and instructors responded 'agree' or 'strongly agree', the VR system was effective, while the opposite indicated that further improvements were needed.

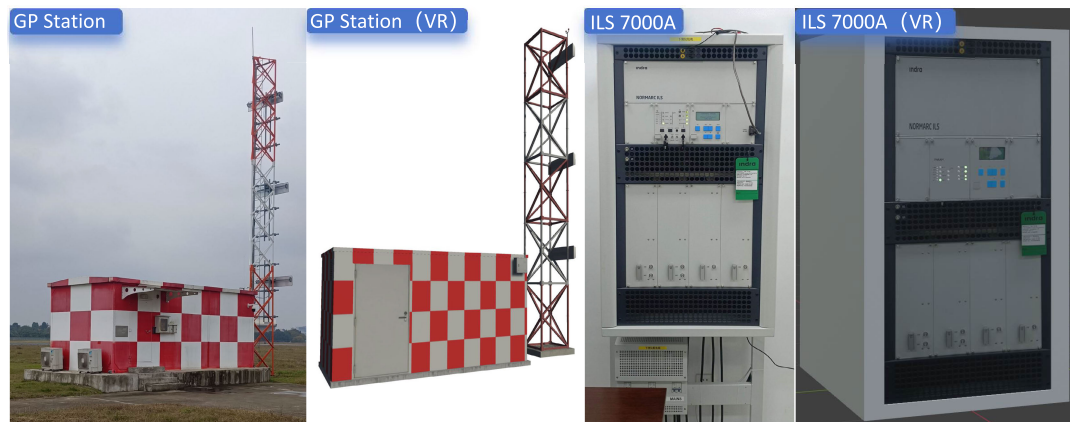


FIGURE 16. ILS 7000A simulation.

The results of the survey are shown in Table 6, and the data show that the participants generally rated DVOR's VSTS positively. Most of them chose 'Agree' and 'Strongly Agree' in the areas of support and assistance, equipment understanding, and troubleshooting skills. Opinions were more mixed, but some were in favour of the statement that using DVOR's VSTS would be enough to troubleshoot real DVOR. In addition to the survey responses, almost all participants expressed a desire for the system to support a wider range of GEAN. Overall, the VR teaching system performed well in terms of improving operational skills and understanding of the equipment, but may require further validation and optimisation for wider application.

The results indicate that all volunteers are satisfied with the system, which demonstrates its practical value. Some volunteers have suggested that the sole focus on DVOR virtual equipment is a limitation of this study. Due to limited research resources and the complexity of equipment fault logic, other GEAN could also theoretically benefit from the application of DT technology.

B. SYSTEM MIGRATION CAPABILITY

1) RTDEHMS MIGRATION CAPABILITY

Architecture suitable for use in scenarios where a DT detection system build is required for GEAN (e.g., DVOR, ILS, DME, NDB). For real-time monitoring of GEAN operational status. Minimum data requirements include the operational status of GEAN (power, frequency, signal strength, modulation system and other key parameter information), data anomalies upper and lower bounds, equipment structural design data (equipment dimensions and other relevant information), and protocol data (explicitly collect the serial communication protocols and data formats of GEAN). Minimum system requirements: with a data collector, capable of serial port data acquisition; with a database and server, capable of storing the operational data of the GEAN; with a virtual entity, capable of receiving real-world data and

presenting it to the user in the virtual world, as well as interacting with it. With mapping capabilities, the system can quickly transfer the collected real-time data to the virtual entity. Satisfying the above minimum data requirements, the minimum requirements of the system can achieve the GEAN's RTDEHMS.

2) VSTS MIGRATION CAPABILITY

Applicable scenario of the architecture: there is a need to learn and train the operation and maintenance of GEAN, but the high price of GEAN and the high requirements for the site environment make it difficult for students to have access to specific types of GEAN.

Minimum data requirements: equipment structure design data (equipment dimensions and other relevant information); equipment normal operation of the parameter range; equipment failure types; equipment failure troubleshooting methods and processes; build equipment failure anomaly logic rules. Failure causes optional data and failure causes fusion is not a necessary condition for fault simulation to be completed.

Minimum requirements of the system: have a virtual model of GEAN, and complete the interactive function through 3D visualisation tools (such as Unity, Unreal Engine, or Web-based Cesium/Three.js). Be able to simulate the changes of the relevant indicator lights and parameters of GEAN when it fails. Be able to carry out troubleshooting through the user's operation. A scoring algorithm that evaluates the user's troubleshooting.

To meet the above minimum data requirements, the minimum system requirements can be achieved GEAN's VSTS.

3) MIGRATION CAPACITY TEST

In order to verify the migration capability of this system architecture, this paper carries out the DT monitoring system construction for the 7000A equipment in the Glide Path (GP) station of ILS system. The steps are as follows.

- Establish a virtual station GP station, as shown in Fig. 16.
- Through the modelling tool Blender to build the virtual entity of the corresponding equipment.
- Write the serial port acquisition algorithm of the corresponding equipment.
- Realise the data mapping and storage of the equipment.
- Through Unity to develop the corresponding RTDEHMS.

The results of the migration capability tests indicate that the system architecture is transferable to other GEAN, laying a solid foundation for the construction of a digital airport.

VIII. CONCLUSION

On the one hand, this paper applies the DT architecture to the construction of GEAN, using the DVOR (VRB-52D) as a case study. RTDEHMS was developed, providing a precise algorithm for data acquisition from the DVOR and enabling bidirectional mapping between the physical and virtual entities. This approach robustly ensures the accuracy and reliability of DVOR operations. Experimental results demonstrate that the proposed method is applicable to a wide range of other GEAN.

RTDEHMS enables real-time data collection and analysis of GEAN, providing timely maintenance recommendations to minimize unplanned downtime, enhance equipment reliability and availability, and optimize lifecycle management while reducing operational and maintenance costs. It ensures the efficiency and precision of GEAN. Furthermore, by continuously accumulating and analyzing operational data, the DT offers robust support for the upgrades and technological iterations of GEAN throughout its entire lifecycle.

The application of DT technology in the aerospace sector faces significant challenges in standardization. The ISO 23247 [61] series of data standards, published by the International Organization for Standardization (ISO) in 2021, provides a general framework for DT in manufacturing, including data structures, information exchange, and operational models. However, it does not address the specific requirements of the aerospace industry. The standardization challenges faced by the aerospace sector primarily stem from the following aspects: data diversity and incompatibility, real-time and high-precision requirements, data security and confidentiality, lifecycle data management, and model interoperability. To address the specific needs of the aerospace sector, many researchers are actively exploring the development of application-specific standards for DT technology [62], [63].

This study also contributes to the standardization and collaboration of GEAN. The proposed RTDEHMS offers practical experience in unifying the modeling, data acquisition, and interface standards for GEAN.

This work promotes the standardization of data and equipment in the field of GEAN, paving the way for enhanced international collaboration. The development of a DT system for GEAN not only significantly improves operational

efficiency and safety but also drives the intelligent [64] and standardized [65] advancement of GEAN. The widespread adoption of such a system will provide critical technological support for the modern aviation industry, ensuring safer and more efficient air traffic operations in the future. Looking ahead, the integration of DT for GEAN with those of aircraft can enable the construction of a comprehensive navigation system ecosystem.

On the other hand, this study also designs and implements VSTS for learning GEAN fault diagnosis. A method for constructing fault logic rules for the GEAN is proposed, which, in combination with the equipment's fault detection procedure, uses fault logic nodes to simulate equipment anomalies. Finally, a strategy for evaluating DVOR fault troubleshooting is presented.

This study contributes to the high quality development of aviation, with significant value to GEAN's technical research and practical training of CNS Personnel. Compared to current practices, The system architecture proposed in this study has the potential to effectively reduce training costs for CNS Personnel, while achieving better training outcomes. VR technology, by creating realistic virtual environments of GEAN, enables trainees to operate, observe, and interact in an immersive setting, greatly enhancing engagement and participation. This approach helps learners gain a more intuitive understanding of the complex structure and working principles of GEAN.

The VR teaching platform can dynamically simulate the operating conditions of GEAN under various scenarios (e.g., signal interference, environmental changes), allowing trainees to learn how the equipment works and how to handle potential issues in a virtual environment. This enhances their understanding and mastery of complex operational processes. GEAN often involves high-voltage, precision instruments, presenting real-world risks. VR teaching offers a zero-risk virtual environment where trainees can practice repeatedly, avoiding damage to real equipment or safety incidents caused by operational errors, thus ensuring both trainee safety and equipment reliability.

Traditional training requires the use of actual GEAN for exercises, which are expensive and complex to deploy. With the VR platform, a wide range of teaching tasks can be completed in a virtual setting, reducing equipment wear and the maintenance costs of on-site training, significantly lowering the time, resources, and financial investment required for training. The VR platform can provide tiered educational content based on the trainee's skill level, ranging from basic knowledge to advanced operations, including equipment maintenance, debugging, and troubleshooting. This satisfies the learning needs of trainees from novice to expert levels.

Furthermore, the VR platform helps trainees master the operational standards and fault-handling methods of GEAN in critical tasks, strengthening their emergency response and troubleshooting abilities. Through virtual simulations of real-world scenarios, trainees can repeatedly practice under

pressure, improving their decision-making and operational skills in complex situations. This contributes to the cultivation of high-quality technical professionals for GEAN and shortens the transition time from training to deployment.

The platform also aligns with the digital transformation trend in aviation, creating a modernized teaching system. Looking ahead, the challenge and opportunity lie in the scalability of the teaching content. The VR platform is not only intended to cover GEAN but must also expand to include teaching content for other critical systems, such as aviation communication and surveillance systems, forming an essential component of digital airports.

Overall, VSTS not only significantly enhances teaching quality but also effectively reduces training costs, improves safety and increases flexibility. It provides strong technological support for the cultivation of high-level professionals in the aviation industry while also driving the modernization and digital transformation of GEAN training.

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